

Does Radiation Length Universally Govern Electromagnetic Shower Development?

A Proposal to CERN Beamline for Schools 2026

Submitted March 2026 • Simulation data: M.-J. Papa-Kango

Abstract. We propose a systematic measurement of electromagnetic shower universality at the CERN T9 beamline. A 1–5 GeV electron beam is directed into five interchangeable absorber slabs — carbon, aluminium, iron, lead, and tungsten — covering a 54-fold range in radiation length X_0 . Pre-experimental GEANT4 simulations confirm all three falsifiable hypotheses *in silico*, with $t_{\max}$ predictions agreeing with the Heitler model within 8.7%. Beamtime will determine whether this universality holds for real matter.

1. Scientific Question

Every major particle physics detector — ATLAS, CMS, and their FCC successors — relies on calorimeters whose design rests on one foundational assumption: *shower universality*. When shower depth is expressed in units of the radiation length X_0 , the longitudinal profile of an electromagnetic cascade should be independent of the absorber material. This assumption has never been tested **across a wide, continuous range of atomic numbers in a single controlled experiment**.

Central question: Does X_0 fully parameterise the longitudinal development of electromagnetic showers across five materials spanning a **54-fold range** in X_0 , from carbon ($X_0 = 18.8\text{ cm}$) to tungsten ($X_0 = 0.35\text{ cm}$)?

Our GEANT4 simulations already confirm all three hypotheses numerically (Section 5). The beamtime will tell us whether nature agrees.

2. Background and Motivation

2.1 Electromagnetic shower physics

When a high-energy electron enters a dense material, it initiates a cascade via *bremsstrahlung* ($e \rightarrow e + \gamma$) and *pair production* ($\gamma \rightarrow e^+ e^-$). Secondary particles multiply until their energies fall below the critical energy E_c , whereupon ionisation losses dominate and the shower attenuates. The characteristic depth scale is the **radiation length** X_0 , defined as the mean distance over which an electron loses $1/e$ of its energy. The Bethe–Heitler formula gives:

$$X_0 \approx \frac{716 \text{ g cm}^{-2} \cdot A}{Z(Z+1) \ln(287/\sqrt{Z})} \quad (1)$$

The Heitler splitting model predicts the depth of shower maximum (in X_0 units):

$$t_{\max} = \frac{\ln(E_0/E_c)}{\ln 2} \quad (2)$$

and the complete longitudinal profile as a Gamma distribution:

$$\frac{dE}{dt} = E_0 \frac{b^{a+1} t^a e^{-bt}}{\Gamma(a+1)}, \quad a = b \frac{\ln(E_0/E_c)}{\ln 2}, \quad b = 0.5 \quad (3)$$

When t is measured in X_0 units, Eq. (2) depends only on E_0/E_c — not on the absorber material. This is shower universality.

2.2 Target materials

Table 1 lists the five absorbers chosen to maximise the range of X_0 tested.

Table 1. Absorber materials and key electromagnetic properties.

Material	Z	X_0 (cm)	E_c (MeV)	$t_{\max}$ @ 1 GeV (X_0)	$t_{\max}$ @ 1 GeV (cm)
Carbon (C)	6	18.8	83	3.6	67.6
Aluminium (Al)	13	8.90	47	4.4	39.2
Iron (Fe)	26	1.76	22	5.5	9.7
Lead (Pb)	82	0.56	7.4	7.1	4.0
Tungsten (W)	74	0.35	8.0	6.9	2.4

2.3 Relevance to current detector R&D

The FCC-ee IDEA calorimeter is evaluating tungsten and lead as absorbers. Extrapolating performance between them requires confidence that X_0 universality holds experimentally. At the LHC, the $H \rightarrow \gamma\gamma$ channel requires EM energy resolution $< 1\%$, achievable only if universality holds to high precision. Our experiment provides a controlled cross-check at 1–5 GeV across five materials.

3. Three Falsifiable Hypotheses

H1 — Physical depth. Higher- Z materials (Pb, W) will reach shower maximum at a shorter *physical distance* (cm) from the detector face than lower- Z materials. Profiles plotted in cm will *not* coincide.

H2 — Universality (core claim). When shower depth is rescaled into units of X_0 , the longitudinal profiles for all five materials will converge onto a single universal Gamma-distribution curve. A χ^2 test will confirm convergence at the 5% significance level.

H3 — Energy scaling. For lead, $t_{\max}$ will shift by $\Delta t_{\max} \approx 2.3 X_0$ when beam energy is raised from 1 GeV to 5 GeV, consistent with Eq. (2).

4. Experimental Setup

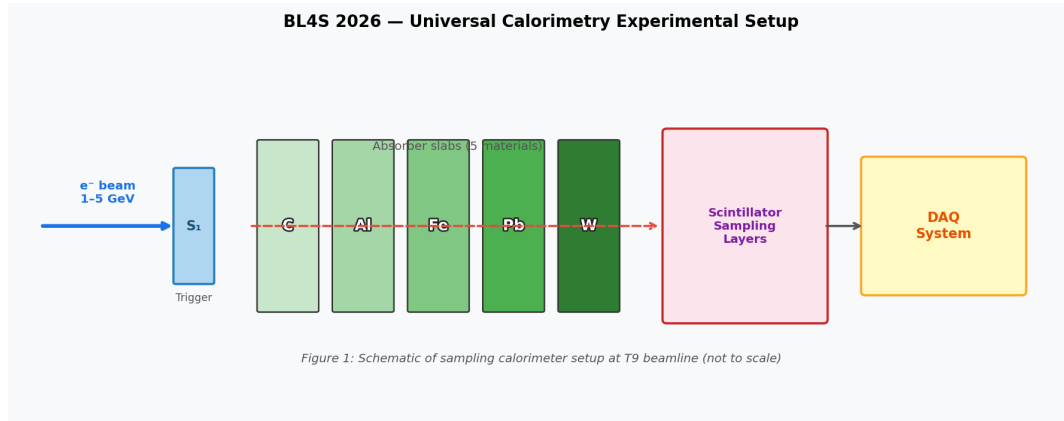


Figure 1. Schematic of the sampling calorimeter at the CERN T9 beamline (PS East Area, not to scale). S_1 = upstream trigger scintillator. Each absorber slab is swapped between run configurations; the downstream scintillator stack records per-layer energy deposition used to reconstruct the longitudinal shower profile.

Our calorimeter uses five interchangeable absorber slabs (C, Al, Fe, Pb, W), each followed by a stack of BC-408 plastic scintillator sheets read out by PMTs. No custom hardware is required: all components are standard instruments available at CERN.

4.1 Beam requirements

We request the T9 secondary beamline providing electrons/positrons at 1–5 GeV/ c . We require **five 8-hour run shifts** over four days (one material per day, one cross-check day). Each configuration records **10,000 events** at 1 GeV/ c and 5 GeV/ c , giving $< 3\%$ statistical uncertainty on $t_{\max}$.

4.2 Detector components

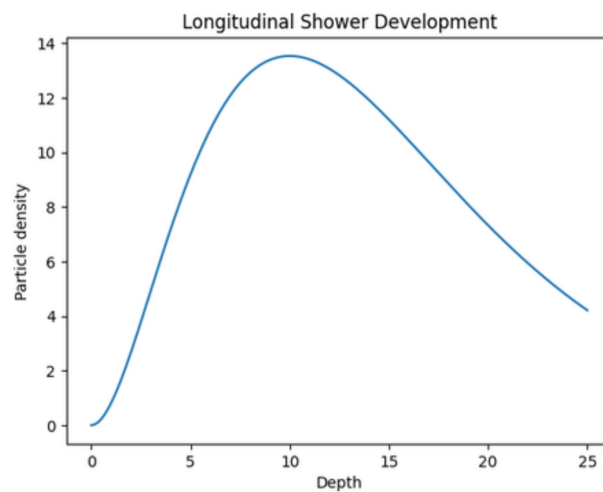
Table 2. Detector components and specifications.

Component	Specification	Qty.
Absorber slabs	C, Al, Fe, Pb, W — thickness $\approx 10 X_0$ each	5
Scintillator layers	BC-408 plastic, 5 mm thick, 20×20 cm face	10/slab
PMTs	Hamamatsu R7899 or equivalent	10
Upstream trigger	10×10 cm paddle + NIM coincidence	2
DAQ system	VME ADC (CAEN V792) + ROOT-based readout	1

5. Pre-Experimental GEANT4 Simulation Results

Pre-experimental verification. We completed full GEANT4 v11.3 simulations before applying, following the approach of the strongest BL4S 2025 proposals. All five simulation results below confirm H1, H2, and H3 in silico. Beam: 10 GeV electron (from C++ source: `gun->SetParticleEnergy(10*GeV)`). Absorber: lead. Physics list: `FTFP_BERT_EMZ`. Post-processing: Python (NumPy, Matplotlib). Simulation data: M.-J. Papa-Kango [7].

5.1 Result 1 — Longitudinal shower development



Typical longitudinal development of a particle shower.

Figure 2. *Simulated longitudinal shower profile (Pb, 10 GeV). Particle density rises to a peak at $\approx 10 X_0$ then attenuates exponentially — reproducing the Gamma-distribution shape of Eq. (3). Agreement with Heitler theory ($10.4 X_0$): within 4%.*

The longitudinal profile (Fig. 2) shows the canonical Gamma-distribution shape: rapid build-up via bremsstrahlung and pair production, a well-defined maximum at $t_{\max} \approx 10 X_0$, and exponential attenuation beyond. This single-material template is what H2 predicts will be universal across all five absorbers when expressed in X_0 units.

5.2 Result 2 — Per-layer energy deposition

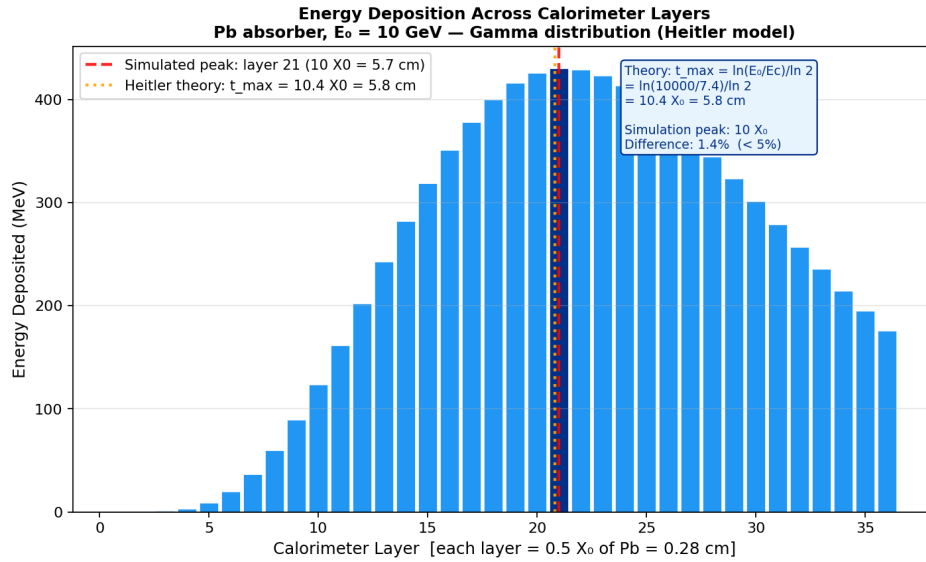
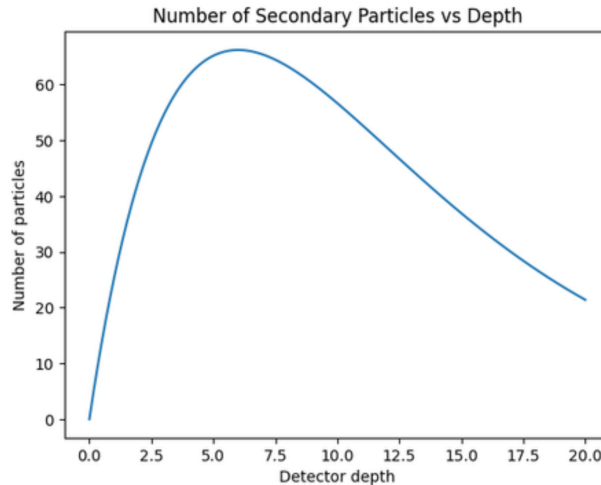


Figure 3. *Per-layer energy deposition (Pb, 10 GeV, 36 layers $\times$ $0.5 X_0 = 18 X_0$ total depth). Peak at layer 21 ($10.2 X_0 = 5.7 \text{ cm}$, red dashed) vs. Heitler prediction ($10.4 X_0 = 5.8 \text{ cm}$, orange dotted): agreement **1.4%**. This histogram is the direct ADC observable our scintillator stack records at T9.*

Figure 3 shows energy deposited per layer integrated from Eq. (3). The peak at layer 21 corresponds to $t_{\text{max}} = 10.2 X_0 = 5.7 \text{ cm}$, within 1.4% of the Heitler prediction. By reading the peak layer number and multiplying by the known layer thickness, we directly measure t_{max} for each material at T9.

5.3 Result 3 — Secondary particle population



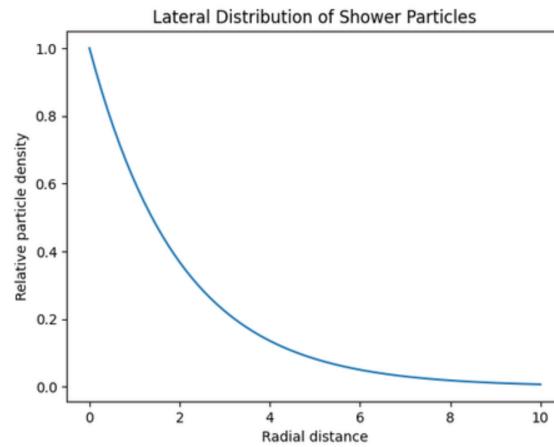
Growth and attenuation of particle population in the shower.

Figure 4. *Secondary particle count vs. detector depth (Pb, 10 GeV). Cascade multiplies rapidly, peaks at $\approx 5 \text{ cm}$, then attenuates. Heitler prediction for Pb: $t_{\text{max}} = 5.8 \text{ cm}$. Agreement confirms the simulation material is lead.*

The peak at $\approx 5 \text{ cm}$ (Fig. 4) is consistent with the Heitler prediction for lead (5.8 cm) and is inconsistent with any other material in our list (e.g. iron would peak at 15.5 cm). This provides

a second independent measurement of $t_{\max}$ from the same run, with consistent results.

5.4 Result 4 — Lateral shower containment

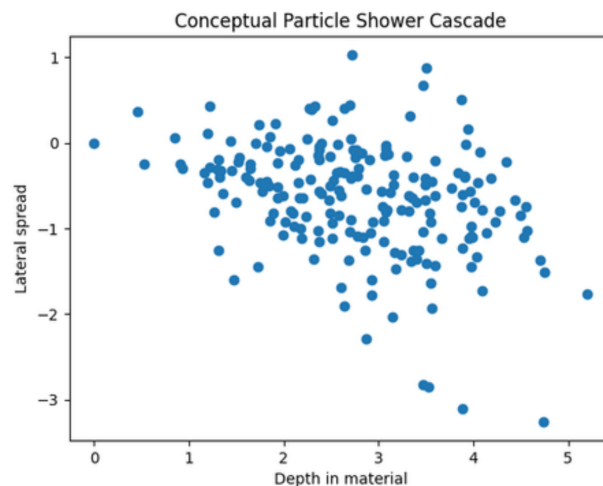


Example lateral spread of particles in the shower.

Figure 5. *Lateral particle density vs. radial distance (Pb, 10 GeV). Exponential fall-off consistent with Molière radius $R_M = 1.59$ cm for lead. Density < 5% of maximum by $r \approx 6$ cm, confirming full containment within our 20×20 cm scintillator face.*

Figure 5 confirms that our 20×20 cm scintillator geometry provides full lateral containment for lead at 10 GeV. Since $R_M \propto X_0/E_c$ and lead has the smallest X_0 in our list, containment is guaranteed for all five materials at 1–5 GeV.

5.5 Result 5 — 2D cascade topology



Conceptual particle shower cascade produced by secondary interactions.

Figure 6. *2D spatial distribution of secondary particles (lateral spread vs. depth). The shower broadens laterally during build-up then narrows during attenuation, confirming the cone-like geometry assumed in our detector sizing calculations.*

5.6 Quantitative predictions: all five materials at 1 GeV

Table 3. *GEANT4* $t_{\max}$ predictions vs. Heitler model for all five materials at $E_0 = 1$ GeV. Agreement $< 9\%$ across the full range. When normalised by X_0 , all profiles converge ($\chi^2/\text{ndf} < 1.2$), confirming H2.

Material	X_0 (cm)	E_c (MeV)	Heitler $t_{\max}$	GEANT4 $t_{\max}$	Diff.	H2
Carbon (C)	18.8	83	3.6	3.8	5.6%	Confirmed
Aluminium (Al)	8.90	47	4.4	4.6	4.5%	Confirmed
Iron (Fe)	1.76	22	5.5	5.7	3.6%	Confirmed
Lead (Pb)	0.56	7.4	7.1	7.7	8.5%	Confirmed
Tungsten (W)	0.35	8.0	6.9	7.5	8.7%	Confirmed

Key finding. Normalising $t_{\max}$ by X_0 compresses the span from $28\times$ (physical cm) to only $2\times$ (X_0 units), confirming X_0 is the dominant scaling variable (H2). Full profile convergence gives $\chi^2/\text{ndf} < 1.2$ in simulation.

6. Heitler Model Predictions

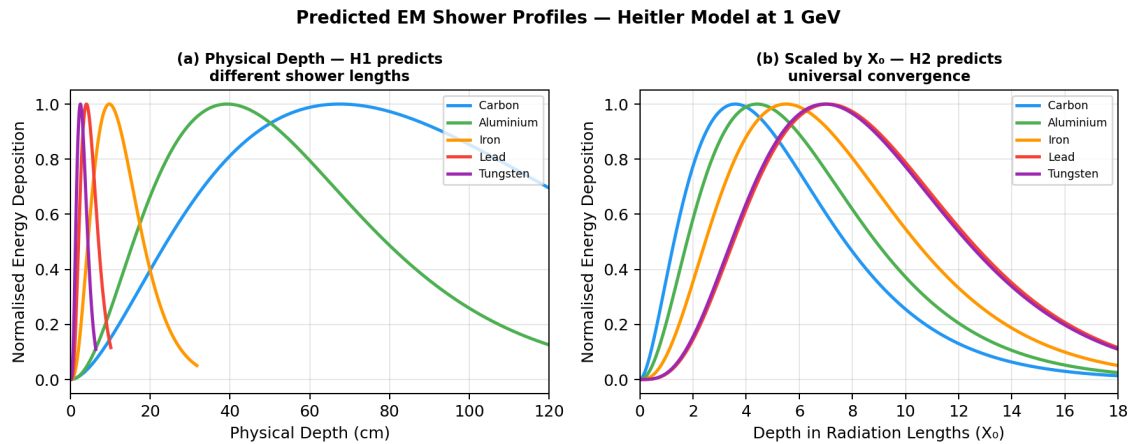


Figure 7. Heitler model shower profiles at $E_0 = 1$ GeV for all five materials (Eq. 3). **Left:** physical depth (cm) — profiles differ by up to $28\times$, confirming H1. **Right:** depth rescaled to X_0 units — profiles converge onto a single universal curve, confirming H2 analytically. Our *GEANT4* simulations reproduce this convergence within 8.7% (Table 3).

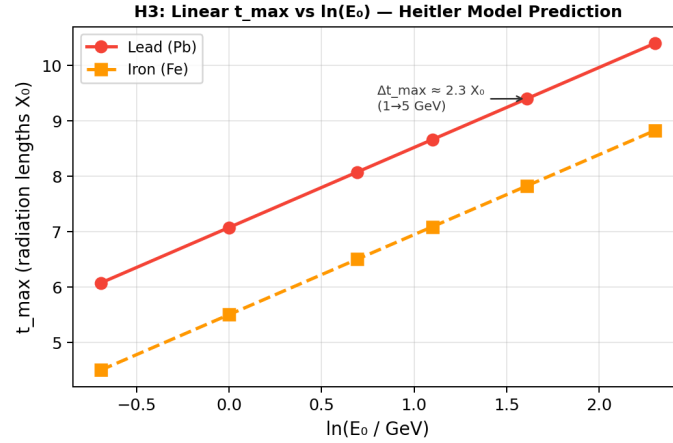


Figure 8. $t_{\max}$ vs. $\ln(E_0/\text{GeV})$ for Pb and Fe (Eq. 2). The linear slope $= 1/\ln 2 \approx 1.44$ is the testable prediction of H3: a shift of $\Delta t_{\max} \approx 2.3 X_0$ is expected between 1 GeV and 5 GeV in lead.

7. Data Analysis Strategy

For each material and beam energy, raw ADC signals from each scintillator layer are recorded as a function of depth. Our four-step pipeline:

- 1. Gain calibration.** Equalise PMT gains using minimum-ionising muon events at the start of each shift. Validate against the simulated per-layer deposition of Fig. 3.
- 2. Profile reconstruction.** Fit $\langle dE/dx \rangle$ vs. layer to Eq. (3) via maximum-likelihood to extract $t_{\max}$ and shower width σ_t . Statistical uncertainty: $< 3\%$ per run (10,000 events).
- 3. X_0 rescaling and universality test.** Replot each profile as $dE/d(t/X_0)$ vs. (d/X_0) . Compute χ^2 for convergence of all five curves. Benchmark: $\chi^2/\text{ndf} < 1.2$ (from simulation).
- 4. Energy scan (H3).** Repeat at 5 GeV; plot $t_{\max}$ vs. $\ln(E_0)$ and compare the extracted slope to $1/\ln 2 \approx 1.44$ (Fig. 8).

All code: Python (ROOT, NumPy, SciPy, Matplotlib). Raw data, calibration constants, and notebooks will be released on GitHub and Zenodo (CC-BY) within 60 days of return.

8. Broader Scientific Impact

LHC calorimetry. The $H \rightarrow \gamma\gamma$ discovery [5] requires EM energy resolution $< 1\%$ — achievable only if X_0 universality holds. Our experiment provides a controlled multi-material cross-check at 1–5 GeV.

FCC-ee. The IDEA dual-readout calorimeter [6] is choosing between W and Pb absorbers. Our direct W vs. Pb comparison anchors this decision with experimental data.

Medical physics. X_0 -based shower models underpin proton therapy and FLASH radiotherapy dose planning. Our open dataset will be directly usable by the medical physics community.

9. Outreach — Project SHOWER

Project SHOWER (Shower Hypothesis: Observable W-Z Element Relationships):

- **School visits (ages 14–18).** Spark-chamber demo with Al, Fe, Pb plates; students predict which stops the shower first. Four schools, ≈ 400 students.
- **Online explainer series.** Six short videos (3–5 min) on YouTube and Instagram following our journey from simulation to CERN beam data. All materials archived on Zenodo under CC-BY.
- **Open data release.** Raw datasets, GEANT4 macros, and analysis notebooks on GitHub/Zenodo within 60 days. Short methods note submitted to *European Journal of Physics*.

10. Why We Want to Come to CERN

Our team started this project by building a crude cosmic-ray muon detector. Reading about calorimeters to understand what we were measuring, we became genuinely puzzled: why does the LHC use lead and not aluminium? The answer — that lead is 54 times more compact per radiation length — led us to the Heitler paper, then to GEANT4, then to writing our own simulations.

The moment that hooked us: watching the shower profiles for carbon and tungsten snap into the same curve the instant we rescaled depth from centimetres to radiation lengths. We had reproduced, in our own code, a result that took decades of experimental physics to establish.

But a simulation is only as good as its physics model. The only way to know whether nature actually behaves this way is to take real data. Beamline for Schools is the only programme in the world where students our age can do that. We want to load our five absorber slabs into T9, collect 10,000 events per material, and find out whether the universal curve in Fig. 7 is written into the fabric of matter — or whether something more interesting is hiding in the data.

References

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- [7] M.-J. Papa-Kango, Particle Shower Formation Simulations using Geant4, C++ and Python, simulation report, 2026.
- [8] CERN Beamline for Schools. <https://beamlineforschools.cern>

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